

Links

person answers each query truthfully, we can find x using $\log n$ queries by successively splitting the sets used in each query in half. Ulam's problem, proposed by Stanislaw Ulam in 1976, asks for the number of queries required to find x , supposing that the first person is allowed to lie exactly once.

- Show that by asking each question twice, given a number x and a set with n elements, and asking one more question when we find the lie, Ulam's problem can be solved using $2 \log n + 1$ queries.
- Show that by dividing the initial set of n elements into four parts, each with $n/4$ elements, $1/4$ of the elements can be eliminated using two queries. [Hint: Use two queries, where each of the queries asks whether the element is in the union of two of the subsets with $n/4$ elements and where one of the subsets of $n/4$ elements is used in both queries.]
- Show from part (b) that if $f(n)$ equals the number of queries used to solve Ulam's problem using the method from part (b) and n is divisible by 4, then $f(n) = f(3n/4) + 2$.
- Solve the recurrence relation in part (c) for $f(n)$.
- Is the naive way to solve Ulam's problem by asking each question twice or the divide-and-conquer method based on part (b) more efficient? The most efficient way to solve Ulam's problem has been determined by A. Pelc [Pe87].

In Exercises 29–33, assume that f is an increasing function satisfying the recurrence relation $f(n) = af(n/b) + cn^d$, where $a \geq 1$, b is an integer greater than 1, and c and d are positive real numbers. These exercises supply a proof of Theorem 2.

- *29. Show that if $a = b^d$ and n is a power of b , then $f(n) = f(1)n^d + cn^d \log_b n$.
30. Use Exercise 29 to show that if $a = b^d$, then $f(n)$ is $O(n^d \log n)$.
- *31. Show that if $a \neq b^d$ and n is a power of b , then $f(n) = C_1 n^d + C_2 n^{\log_b a}$, where $C_1 = b^d c / (b^d - a)$ and $C_2 = f(1) + b^d c / (a - b^d)$.
32. Use Exercise 31 to show that if $a < b^d$, then $f(n)$ is $O(n^d)$.
33. Use Exercise 31 to show that if $a > b^d$, then $f(n)$ is $O(n^{\log_b a})$.
34. Find $f(n)$ when $n = 4^k$, where f satisfies the recurrence relation $f(n) = 5f(n/4) + 6n$, with $f(1) = 1$.
35. Give a big- O estimate for the function f in Exercise 34 if f is an increasing function.
36. Find $f(n)$ when $n = 2^k$, where f satisfies the recurrence relation $f(n) = 8f(n/2) + n^2$ with $f(1) = 1$.
37. Give a big- O estimate for the function f in Exercise 36 if f is an increasing function.

8.4 Generating Functions

8.4.1 Introduction

Links

Generating functions are used to represent sequences efficiently by coding the terms of a sequence as coefficients of powers of a variable x in a formal power series. Generating functions can be used to solve many types of counting problems, such as the number of ways to select or distribute objects of different kinds, subject to a variety of constraints, and the number of ways to make change for a dollar using coins of different denominations. Generating functions can be used to solve recurrence relations by translating a recurrence relation for the terms of a sequence into an equation involving a generating function. This equation can then be solved to find a closed form for the generating function. From this closed form, the coefficients of the power series for the generating function can be found, solving the original recurrence relation. Generating functions can also be used to prove combinatorial identities by taking advantage of relatively simple relationships between functions that can be translated into identities involving the terms of sequences. Generating functions are a helpful tool for studying many properties of sequences besides those described in this section, such as their use for establishing asymptotic formulae for the terms of a sequence.

We begin with the definition of the generating function for a sequence.

Definition 1

The *generating function for the sequence* $a_0, a_1, \dots, a_k, \dots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k.$$

Remark: The generating function for $\{a_k\}$ given in Definition 1 is sometimes called the **ordinary generating function** of $\{a_k\}$ to distinguish it from other types of generating functions for this sequence.

EXAMPLE 1 The generating functions for the sequences $\{a_k\}$ with $a_k = 3$, $a_k = k + 1$, and $a_k = 2^k$ are $\sum_{k=0}^{\infty} 3x^k$, $\sum_{k=0}^{\infty} (k+1)x^k$, and $\sum_{k=0}^{\infty} 2^k x^k$, respectively. 

Extra Examples 

We can define generating functions for finite sequences of real numbers by extending a finite sequence $a_0, a_1, \dots, a_n$ into an infinite sequence by setting $a_{n+1} = 0$, $a_{n+2} = 0$, and so on. The generating function $G(x)$ of this infinite sequence $\{a_n\}$ is a polynomial of degree n because no terms of the form $a_j x^j$ with $j > n$ occur, that is,

$$G(x) = a_0 + a_1 x + \cdots + a_n x^n.$$

EXAMPLE 2 What is the generating function for the sequence 1, 1, 1, 1, 1, 1?

Solution: The generating function of 1, 1, 1, 1, 1, 1 is

$$1 + x + x^2 + x^3 + x^4 + x^5.$$

By Theorem 1 of Section 2.4 we have

$$(x^6 - 1)/(x - 1) = 1 + x + x^2 + x^3 + x^4 + x^5$$

when $x \neq 1$. Consequently, $G(x) = (x^6 - 1)/(x - 1)$ is the generating function of the sequence 1, 1, 1, 1, 1, 1. [Because the powers of x are only place holders for the terms of the sequence in a generating function, we do not need to worry that $G(1)$ is undefined.] 

EXAMPLE 3 Let m be a positive integer. Let $a_k = C(m, k)$, for $k = 0, 1, 2, \dots, m$. What is the generating function for the sequence $a_0, a_1, \dots, a_m$?

Solution: The generating function for this sequence is

$$G(x) = C(m, 0) + C(m, 1)x + C(m, 2)x^2 + \cdots + C(m, m)x^m.$$

The binomial theorem shows that $G(x) = (1 + x)^m$. 

8.4.2 Useful Facts About Power Series

When generating functions are used to solve counting problems, they are usually considered to be **formal power series**. As such, they are treated as algebraic objects; questions about their convergence are ignored. However, when formal power series are convergent, valid operations carry over to their use as formal power series. We will take advantage of the power series of particular functions around $x = 0$. These power series are unique and have a positive radius of convergence. Readers familiar with calculus can consult textbooks on this subject for details about power series, including the convergence of the series we use here.

We will now state some widely important facts about infinite series used when working with generating functions. These facts can be found in calculus texts.

EXAMPLE 4 The function $f(x) = 1/(1 - x)$ is the generating function of the sequence 1, 1, 1, 1, ..., because

$$1/(1 - x) = 1 + x + x^2 + \dots$$

for $|x| < 1$. 

EXAMPLE 5 The function $f(x) = 1/(1 - ax)$ is the generating function of the sequence 1, a , a^2 , a^3 , ..., because

$$1/(1 - ax) = 1 + ax + a^2x^2 + \dots$$

when $|ax| < 1$, or equivalently, for $|x| < 1/|a|$ for $a \neq 0$. 

We also will need some results on how to add and how to multiply two generating functions. Proofs of these results can be found in calculus texts.

THEOREM 1

Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$. Then

$$f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k \quad \text{and} \quad f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k.$$

Remark: Theorem 1 is valid only for power series that converge in an interval, as all series considered in this section do. However, the theory of generating functions is not limited to such series. In the case of series that do not converge, the statements in Theorem 1 can be taken as definitions of addition and multiplication of generating functions.

We will illustrate how Theorem 1 can be used with Example 6.

EXAMPLE 6 Let $f(x) = 1/(1 - x)^2$. Use Example 4 to find the coefficients $a_0, a_1, a_2, \dots$ in the expansion $f(x) = \sum_{k=0}^{\infty} a_k x^k$.

Solution: From Example 4 we see that

$$1/(1 - x) = 1 + x + x^2 + x^3 + \dots$$

Hence, from Theorem 1, we have

$$1/(1 - x)^2 = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k 1 \right) x^k = \sum_{k=0}^{\infty} (k + 1) x^k.$$


Remark: This result also can be derived from Example 4 by differentiation. Taking derivatives is a useful technique for producing new identities from existing identities for generating functions.

To use generating functions to solve many important counting problems, we will need to apply the binomial theorem for exponents that are not positive integers. Before we state an extended version of the binomial theorem, we need to define extended binomial coefficients.

Definition 2

Let u be a real number and k a nonnegative integer. Then the *extended binomial coefficient* $\binom{u}{k}$ is defined by

$$\binom{u}{k} = \begin{cases} u(u-1) \cdots (u-k+1)/k! & \text{if } k > 0, \\ 1 & \text{if } k = 0. \end{cases}$$

EXAMPLE 7 Find the values of the extended binomial coefficients $\binom{-2}{3}$ and $\binom{1/2}{3}$.

Solution: Taking $u = -2$ and $k = 3$ in Definition 2 gives us

$$\binom{-2}{3} = \frac{(-2)(-3)(-4)}{3!} = -4.$$

Similarly, taking $u = 1/2$ and $k = 3$ gives us

$$\begin{aligned} \binom{1/2}{3} &= \frac{(1/2)(1/2-1)(1/2-2)}{3!} \\ &= (1/2)(-1/2)(-3/2)/6 \\ &= 1/16. \end{aligned}$$

Example 8 provides a useful formula for extended binomial coefficients when the top parameter is a negative integer. It will be useful in our subsequent discussions.

EXAMPLE 8 When the top parameter is a negative integer, the extended binomial coefficient can be expressed in terms of an ordinary binomial coefficient. To see that this is the case, note that

$$\begin{aligned} \binom{-n}{r} &= \frac{(-n)(-n-1) \cdots (-n-r+1)}{r!} && \text{by definition of extended binomial coefficient} \\ &= \frac{(-1)^r n(n+1) \cdots (n+r-1)}{r!} && \text{factoring out } -1 \text{ from each term in the numerator} \\ &= \frac{(-1)^r (n+r-1)(n+r-2) \cdots n}{r!} && \text{by the commutative law for multiplication} \\ &= \frac{(-1)^r (n+r-1)!}{r!(n-1)!} && \text{multiplying both the numerator and denominator by } (n-1)! \\ &= (-1)^r \binom{n+r-1}{r} && \text{by the definition of binomial coefficients} \\ &= (-1)^r C(n+r-1, r) && \text{using alternative notation for binomial coefficients.} \end{aligned}$$

We now state the extended binomial theorem.

THEOREM 2 THE EXTENDED BINOMIAL THEOREM Let x be a real number with $|x| < 1$ and let u be a real number. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k.$$

Theorem 2 can be proved using the theory of Maclaurin series. We leave its proof to the reader with a familiarity with this part of calculus.

Remark: When u is a positive integer, the extended binomial theorem reduces to the binomial theorem presented in Section 6.4, because in that case $\binom{u}{k} = 0$ if $k > u$.

Example 9 illustrates the use of Theorem 2 when the exponent is a negative integer.

EXAMPLE 9 Find the generating functions for $(1+x)^{-n}$ and $(1-x)^{-n}$, where n is a positive integer, using the extended binomial theorem.

Solution: By the extended binomial theorem, it follows that

$$(1+x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k.$$

Using Example 8, which provides a simple formula for $\binom{-n}{k}$, we obtain

$$(1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) x^k.$$

Replacing x by $-x$, we find that

$$(1-x)^{-n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k.$$

Table 1 presents a useful summary of some generating functions that arise frequently.

Remark: Note that the second and third formulae in this table can be deduced from the first formula by substituting ax and x^r for x , respectively. Similarly, the sixth and seventh formulae can be deduced from the fifth formula using the same substitutions. The tenth and eleventh can be deduced from the ninth formula by substituting $-x$ and ax for x , respectively. Also, some of the formulae in this table can be derived from other formulae using methods from calculus (such as differentiation and integration). Students are encouraged to know the core formulae in this table (that is, formulae from which the others can be derived, perhaps the first, fourth, fifth, eighth, ninth, twelfth, and thirteenth formulae) and understand how to derive the other formulae from these core formulae.

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \cdots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \cdots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \cdots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \cdots$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \cdots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2x^2 + \cdots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1)^{k+1}/k$

Note: The series for the last two generating functions can be found in most calculus books when power series are discussed.

8.4.3 Counting Problems and Generating Functions

Generating functions can be used to solve a wide variety of counting problems. In particular, they can be used to count the number of combinations of various types. In Chapter 6 we developed techniques to count the r -combinations from a set with n elements when repetition is allowed and additional constraints may exist. Such problems are equivalent to counting the solutions to equations of the form

$$e_1 + e_2 + \cdots + e_n = C,$$

where C is a constant and each e_i is a nonnegative integer that may be subject to a specified constraint. Generating functions can also be used to solve counting problems of this type, as Examples 10–12 show.

EXAMPLE 10 Find the number of solutions of

*Extra
Examples* 

$$e_1 + e_2 + e_3 = 17,$$

where e_1, e_2 , and e_3 are nonnegative integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

Solution: The number of solutions with the indicated constraints is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7).$$

This follows because we obtain a term equal to x^{17} in the product by picking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3} , where the exponents e_1, e_2 , and e_3 satisfy the equation $e_1 + e_2 + e_3 = 17$ and the given constraints.

It is not hard to see that the coefficient of x^{17} in this product is 3. Hence, there are three solutions. (Note that the calculating of this coefficient involves about as much work as enumerating all the solutions of the equation with the given constraints. However, the method that this illustrates often can be used to solve wide classes of counting problems with special formulae, as we will see. Furthermore, a computer algebra system can be used to do such computations.) 

EXAMPLE 11 In how many different ways can eight identical cookies be distributed among three distinct children if each child receives at least two cookies and no more than four cookies?

Solution: Because each child receives at least two but no more than four cookies, for each child there is a factor equal to

$$(x^2 + x^3 + x^4)$$

in the generating function for the sequence $\{c_n\}$, where c_n is the number of ways to distribute n cookies. Because there are three children, this generating function is

$$(x^2 + x^3 + x^4)^3.$$

We need the coefficient of x^8 in this product. The reason is that the x^8 terms in the expansion correspond to the ways that three terms can be selected, with one from each factor, that have exponents adding up to 8. Furthermore, the exponents of the term from the first, second, and third factors are the numbers of cookies the first, second, and third children receive, respectively. Computation shows that this coefficient equals 6. Hence, there are six ways to distribute the cookies so that each child receives at least two, but no more than four, cookies. 

EXAMPLE 12 Use generating functions to determine the number of ways to insert tokens worth \$1, \$2, and \$5 into a vending machine to pay for an item that costs r dollars in both the cases when the order in which the tokens are inserted does not matter and when the order does matter. (For example, there are two ways to pay for an item that costs \$3 when the order in which the tokens are inserted does not matter: inserting three \$1 tokens or one \$1 token and a \$2 token. When the order matters, there are three ways: inserting three \$1 tokens, inserting a \$1 token and then a \$2 token, or inserting a \$2 token and then a \$1 token.)

Solution: Consider the case when the order in which the tokens are inserted does not matter. Here, all we care about is the number of each token used to produce a total of r dollars. Because we can use any number of \$1 tokens, any number of \$2 tokens, and any number of \$5 tokens, the answer is the coefficient of x^r in the generating function

$$(1 + x + x^2 + x^3 + \cdots)(1 + x^2 + x^4 + x^6 + \cdots)(1 + x^5 + x^{10} + x^{15} + \cdots).$$

(The first factor in this product represents the \$1 tokens used, the second the \$2 tokens used, and the third the \$5 tokens used.) For example, the number of ways to pay for an item costing \$7 using \$1, \$2, and \$5 tokens is given by the coefficient of x^7 in this expansion, which equals 6.

When the order in which the tokens are inserted matters, the number of ways to insert exactly n tokens to produce a total of r dollars is the coefficient of x^r in

$$(x + x^2 + x^5)^n,$$

because each of the r tokens may be a \$1 token, a \$2 token, or a \$5 token. Because any number of tokens may be inserted, the number of ways to produce r dollars using \$1, \$2, or \$5 tokens, when the order in which the tokens are inserted matters, is the coefficient of x^r in

$$\begin{aligned} 1 + (x + x^2 + x^5) + (x + x^2 + x^5)^2 + \cdots &= \frac{1}{1 - (x + x^2 + x^5)} \\ &= \frac{1}{1 - x - x^2 - x^5}, \end{aligned}$$

where we have added the number of ways to insert 0 tokens, 1 token, 2 tokens, 3 tokens, and so on, and where we have used the identity $1/(1 - x) = 1 + x + x^2 + \cdots$ with x replaced with $x + x^2 + x^5$. For example, the number of ways to pay for an item costing \$7 using \$1, \$2, and \$5 tokens, when the order in which the tokens are used matters, is the coefficient of x^7 in this expansion, which equals 26. [Hint: To see that this coefficient equals 26 requires the addition of the coefficients of x^7 in the expansions $(x + x^2 + x^5)^k$ for $2 \leq k \leq 7$. This can be done by hand with considerable computation, or a computer algebra system can be used.] ◀

Example 13 shows the versatility of generating functions when used to solve problems with differing assumptions.

EXAMPLE 13 Use generating functions to find the number of k -combinations of a set with n elements. Assume that the binomial theorem has already been established.

Solution: Each of the n elements in the set contributes the term $(1 + x)$ to the generating function $f(x) = \sum_{k=0}^n a_k x^k$. Here $f(x)$ is the generating function for $\{a_k\}$, where a_k represents the number of k -combinations of a set with n elements. Hence,

$$f(x) = (1 + x)^n.$$

But by the binomial theorem, we have

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k,$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Hence, $C(n, k)$, the number of k -combinations of a set with n elements, is

$$\frac{n!}{k!(n-k)!}.$$

Remark: We proved the binomial theorem in Section 6.4 using the formula for the number of r -combinations of a set with n elements. This example shows that the binomial theorem, which can be proved by mathematical induction, can be used to derive the formula for the number of r -combinations of a set with n elements.

EXAMPLE 14 Use generating functions to find the number of r -combinations from a set with n elements when repetition of elements is allowed.

Solution: Let $G(x)$ be the generating function for the sequence $\{a_r\}$, where a_r equals the number of r -combinations of a set with n elements with repetitions allowed. That is, $G(x) = \sum_{r=0}^{\infty} a_r x^r$. Because we can select any number of a particular member of the set with n elements when we form an r -combination with repetition allowed, each of the n elements contributes $(1 + x + x^2 + x^3 + \cdots)$ to a product expansion for $G(x)$. Each element contributes this factor because it may be selected zero times, one time, two times, three times, and so on, when an r -combination is formed (with a total of r elements selected). Because there are n elements in the set and each contributes this same factor to $G(x)$, we have

$$G(x) = (1 + x + x^2 + \cdots)^n.$$

As long as $|x| < 1$, we have $1 + x + x^2 + \cdots = 1/(1 - x)$, so

$$G(x) = 1/(1 - x)^n = (1 - x)^{-n}.$$

Applying the extended binomial theorem (Theorem 2), it follows that

$$(1 - x)^{-n} = (1 + (-x))^{-n} = \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r.$$

The number of r -combinations of a set with n elements with repetitions allowed, when r is a positive integer, is the coefficient a_r of x^r in this sum. Consequently, using Example 8 we find that a_r equals

$$\begin{aligned} \binom{-n}{r} (-1)^r &= (-1)^r C(n + r - 1, r) \cdot (-1)^r \\ &= C(n + r - 1, r). \end{aligned}$$

Note that the result in Example 14 is the same result we stated as Theorem 2 in Section 6.5.

EXAMPLE 15 Use generating functions to find the number of ways to select r objects of n different kinds if we must select at least one object of each kind.

Solution: Because we need to select at least one object of each kind, each of the n kinds of objects contributes the factor $(x + x^2 + x^3 + \cdots)$ to the generating function $G(x)$ for the sequence $\{a_r\}$, where a_r is the number of ways to select r objects of n different kinds if we need at least one object of each kind. Hence,

$$G(x) = (x + x^2 + x^3 + \cdots)^n = x^n(1 + x + x^2 + \cdots)^n = x^n/(1 - x)^n.$$

Using the extended binomial theorem and Example 8, we have

$$\begin{aligned} G(x) &= x^n/(1 - x)^n \\ &= x^n \cdot (1 - x)^{-n} \\ &= x^n \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r \\ &= x^n \sum_{r=0}^{\infty} (-1)^r C(n + r - 1, r) (-1)^r x^r \\ &= \sum_{r=0}^{\infty} C(n + r - 1, r) x^{n+r} \\ &= \sum_{t=n}^{\infty} C(t - 1, t - n) x^t \\ &= \sum_{r=n}^{\infty} C(r - 1, r - n) x^r. \end{aligned}$$

We have shifted the summation in the next-to-last equality by setting $t = n + r$ so that $t = n$ when $r = 0$ and $n + r - 1 = t - 1$, and then we replaced t by r as the index of summation in the last equality to return to our original notation. Hence, there are $C(r - 1, r - n)$ ways to select r objects of n different kinds if we must select at least one object of each kind. 

8.4.4 Using Generating Functions to Solve Recurrence Relations

We can find the solution to a recurrence relation and its initial conditions by finding an explicit formula for the associated generating function. This is illustrated in Examples 16 and 17.

EXAMPLE 16 Solve the recurrence relation $a_k = 3a_{k-1}$ for $k = 1, 2, 3, \dots$ and initial condition $a_0 = 2$.

Extra Examples 

Solution: Let $G(x)$ be the generating function for the sequence $\{a_k\}$, that is, $G(x) = \sum_{k=0}^{\infty} a_k x^k$. First note that

$$xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k.$$

Using the recurrence relation, we see that

$$\begin{aligned}
 G(x) - 3xG(x) &= \sum_{k=0}^{\infty} a_k x^k - 3 \sum_{k=1}^{\infty} a_{k-1} x^k \\
 &= a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1}) x^k \\
 &= 2,
 \end{aligned}$$

because $a_0 = 2$ and $a_k = 3a_{k-1}$. Thus,

$$G(x) - 3xG(x) = (1 - 3x)G(x) = 2.$$

Solving for $G(x)$ shows that $G(x) = 2/(1 - 3x)$. Using the identity $1/(1 - ax) = \sum_{k=0}^{\infty} a^k x^k$, from Table 1, we have

$$G(x) = 2 \sum_{k=0}^{\infty} 3^k x^k = \sum_{k=0}^{\infty} 2 \cdot 3^k x^k.$$

Consequently, $a_k = 2 \cdot 3^k$. 

EXAMPLE 17 Suppose that a valid codeword is an n -digit number in decimal notation containing an even number of 0s. Let a_n denote the number of valid codewords of length n . In Example 4 of Section 8.1 we showed that the sequence $\{a_n\}$ satisfies the recurrence relation

$$a_n = 8a_{n-1} + 10^{n-1}$$

and the initial condition $a_1 = 9$. Use generating functions to find an explicit formula for a_n .

Solution: To make our work with generating functions simpler, we extend this sequence by setting $a_0 = 1$; when we assign this value to a_0 and use the recurrence relation, we have $a_1 = 8a_0 + 10^0 = 8 + 1 = 9$, which is consistent with our original initial condition. (It also makes sense because there is one code word of length 0—the empty string.)

We multiply both sides of the recurrence relation by x^n to obtain

$$a_n x^n = 8a_{n-1} x^n + 10^{n-1} x^n.$$

Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$ be the generating function of the sequence $a_0, a_1, a_2, \dots$. We sum both sides of the last equation starting with $n = 1$, to find that

$$\begin{aligned}
 G(x) - 1 &= \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} x^n + 10^{n-1} x^n) \\
 &= 8 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n \\
 &= 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1} \\
 &= 8x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} 10^n x^n \\
 &= 8xG(x) + x/(1 - 10x),
 \end{aligned}$$

where we have used Example 5 to evaluate the second summation. Therefore, we have

$$G(x) - 1 = 8xG(x) + x/(1 - 10x).$$

Solving for $G(x)$ shows that

$$G(x) = \frac{1 - 9x}{(1 - 8x)(1 - 10x)}.$$

Expanding the right-hand side of this equation into partial fractions (as is done in the integration of rational functions studied in calculus) gives

$$G(x) = \frac{1}{2} \left(\frac{1}{1 - 8x} + \frac{1}{1 - 10x} \right).$$

Using Example 5 twice (once with $a = 8$ and once with $a = 10$) gives

$$\begin{aligned} G(x) &= \frac{1}{2} \left(\sum_{n=0}^{\infty} 8^n x^n + \sum_{n=0}^{\infty} 10^n x^n \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{2} (8^n + 10^n) x^n. \end{aligned}$$

Consequently, we have shown that

$$a_n = \frac{1}{2} (8^n + 10^n).$$



8.4.5 Proving Identities via Generating Functions

In Chapter 6 we saw how combinatorial identities could be established using combinatorial proofs. Here we will show that such identities, as well as identities for extended binomial coefficients, can be proved using generating functions. Sometimes the generating function approach is simpler than other approaches, especially when it is simpler to work with the closed form of a generating function than with the terms of the sequence themselves. We illustrate how generating functions can be used to prove identities with Example 18.

EXAMPLE 18 Use generating functions to show that

$$\sum_{k=0}^n C(n, k)^2 = C(2n, n)$$

whenever n is a positive integer.

Solution: First note that by the binomial theorem $C(2n, n)$ is the coefficient of x^n in $(1 + x)^{2n}$. However, we also have

$$\begin{aligned} (1 + x)^{2n} &= [(1 + x)^n]^2 \\ &= [C(n, 0) + C(n, 1)x + C(n, 2)x^2 + \cdots + C(n, n)x^n]^2. \end{aligned}$$

The coefficient of x^n in this expression is

$$C(n, 0)C(n, n) + C(n, 1)C(n, n-1) + C(n, 2)C(n, n-2) + \cdots + C(n, n)C(n, 0).$$

This equals $\sum_{k=0}^n C(n, k)^2$, because $C(n, n-k) = C(n, k)$. Because both $C(2n, n)$ and $\sum_{k=0}^n C(n, k)^2$ represent the coefficient of x^n in $(1+x)^{2n}$, they must be equal. 

Exercises 44 and 45 ask that Pascal's identity and Vandermonde's identity be proved using generating functions.

Exercises

- Find the generating function for the finite sequence 2, 2, 2, 2, 2.
- Find the generating function for the finite sequence 1, 4, 16, 64, 256.

In Exercises 3–8, by a **closed form** we mean an algebraic expression not involving a summation over a range of values or the use of ellipses.

- Find a closed form for the generating function for each of these sequences. (For each sequence, use the most obvious choice of a sequence that follows the pattern of the initial terms listed.)
 - 0, 2, 2, 2, 2, 2, 0, 0, 0, 0, 0, ...
 - 0, 0, 0, 1, 1, 1, 1, 1, ...
 - 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, ...
 - 2, 4, 8, 16, 32, 64, 128, 256, ...
 - $\binom{7}{0}, \binom{7}{1}, \binom{7}{2}, \dots, \binom{7}{7}, 0, 0, 0, 0, 0, \dots$
 - 2, -2, 2, -2, 2, -2, 2, -2, ...
 - 1, 1, 0, 1, 1, 1, 1, 1, 1, ...
 - 0, 0, 0, 1, 2, 3, 4, ...
- Find a closed form for the generating function for each of these sequences. (Assume a general form for the terms of the sequence, using the most obvious choice of such a sequence.)
 - 1, -1, -1, -1, -1, -1, -1, 0, 0, 0, 0, 0, ...
 - 1, 3, 9, 27, 81, 243, 729, ...
 - 0, 0, 3, -3, 3, -3, 3, -3, ...
 - 1, 2, 1, 1, 1, 1, 1, 1, ...
 - $\binom{7}{0}, 2\binom{7}{1}, 2^2\binom{7}{2}, \dots, 2^7\binom{7}{7}, 0, 0, 0, 0, \dots$
 - 3, 3, -3, 3, -3, 3, ...
 - 0, 1, -2, 4, -8, 16, -32, 64, ...
 - 1, 0, 1, 0, 1, 0, 1, 0, ...
- Find a closed form for the generating function for the sequence $\{a_n\}$, where
 - $a_n = 5$ for all $n = 0, 1, 2, \dots$
 - $a_n = 3^n$ for all $n = 0, 1, 2, \dots$
 - $a_n = 2$ for $n = 3, 4, 5, \dots$ and $a_0 = a_1 = a_2 = 0$.
 - $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$

$$\text{e) } a_n = \binom{8}{n} \text{ for all } n = 0, 1, 2, \dots$$

$$\text{f) } a_n = \binom{n+4}{n} \text{ for all } n = 0, 1, 2, \dots$$

- Find a closed form for the generating function for the sequence $\{a_n\}$, where
 - $a_n = -1$ for all $n = 0, 1, 2, \dots$
 - $a_n = 2^n$ for $n = 1, 2, 3, 4, \dots$ and $a_0 = 0$.
 - $a_n = n - 1$ for $n = 0, 1, 2, \dots$
 - $a_n = 1/(n+1)!$ for $n = 0, 1, 2, \dots$
 - $a_n = \binom{n}{2}$ for $n = 0, 1, 2, \dots$
 - $a_n = \binom{10}{n+1}$ for $n = 0, 1, 2, \dots$
- For each of these generating functions, provide a closed formula for the sequence it determines.
 - $(3x-4)^3$
 - $(x^3+1)^3$
 - $1/(1-5x)$
 - $x^3/(1+3x)$
 - $x^2+3x+7+(1/(1-x^2))$
 - $(x^4/(1-x^4)) - x^3 - x^2 - x - 1$
 - $x^2/(1-x)^2$
 - $2e^{2x}$
- For each of these generating functions, provide a closed formula for the sequence it determines.
 - $(x^2+1)^3$
 - $(3x-1)^3$
 - $1/(1-2x^2)$
 - $x^2/(1-x)^3$
 - $x-1+(1/(1-3x))$
 - $(1+x^3)/(1+x)^3$
 - $x/(1+x+x^2)$
 - $e^{3x^2}-1$
- Find the coefficient of x^{10} in the power series of each of these functions.
 - $(1+x^5+x^{10}+x^{15}+\dots)^3$
 - $(x^3+x^4+x^5+x^6+x^7+\dots)^3$
 - $(x^4+x^5+x^6)(x^3+x^4+x^5+x^6+x^7)(1+x+x^2+x^3+x^4+\dots)$
 - $(x^2+x^4+x^6+x^8+\dots)(x^3+x^6+x^9+\dots)(x^4+x^8+x^{12}+\dots)$
 - $(1+x^2+x^4+x^6+x^8+\dots)(1+x^4+x^8+x^{12}+\dots)(1+x^6+x^{12}+x^{18}+\dots)$
- Find the coefficient of x^9 in the power series of each of these functions.
 - $(1+x^3+x^6+x^9+\dots)^3$
 - $(x^2+x^3+x^4+x^5+x^6+\dots)^3$
 - $(x^3+x^5+x^6)(x^3+x^4)(x+x^2+x^3+x^4+\dots)$
 - $(x+x^4+x^7+x^{10}+\dots)(x^2+x^4+x^6+x^8+\dots)$
 - $(1+x+x^2)^3$

11. Find the coefficient of x^{10} in the power series of each of these functions.
 - a) $1/(1-2x)$
 - b) $1/(1+x)^2$
 - c) $1/(1-x)^3$
 - d) $1/(1+2x)^4$
 - e) $x^4/(1-3x)^3$
12. Find the coefficient of x^{12} in the power series of each of these functions.
 - a) $1/(1+3x)$
 - b) $1/(1-2x)^2$
 - c) $1/(1+x)^8$
 - d) $1/(1-4x)^3$
 - e) $x^3/(1+4x)^2$
13. Use generating functions to determine the number of different ways 10 identical balloons can be given to four children if each child receives at least two balloons.
14. Use generating functions to determine the number of different ways 12 identical action figures can be given to five children so that each child receives at most three action figures.
15. Use generating functions to determine the number of different ways 15 identical stuffed animals can be given to six children so that each child receives at least one but no more than three stuffed animals.
16. Use generating functions to find the number of ways to choose a dozen bagels from three varieties—egg, salty, and plain—if at least two bagels of each kind but no more than three salty bagels are chosen.
17. In how many ways can 25 identical donuts be distributed to four police officers so that each officer gets at least three but no more than seven donuts?
18. Use generating functions to find the number of ways to select 14 balls from a jar containing 100 red balls, 100 blue balls, and 100 green balls so that no fewer than 3 and no more than 10 blue balls are selected. Assume that the order in which the balls are drawn does not matter.
19. What is the generating function for the sequence $\{c_k\}$, where c_k is the number of ways to make change for k dollars using \$1 bills, \$2 bills, \$5 bills, and \$10 bills?
20. What is the generating function for the sequence $\{c_k\}$, where c_k represents the number of ways to make change for k pesos using bills worth 10 pesos, 20 pesos, 50 pesos, and 100 pesos?
21. Give a combinatorial interpretation of the coefficient of x^4 in the expansion $(1+x+x^2+x^3+\cdots)^3$. Use this interpretation to find this number.
22. Give a combinatorial interpretation of the coefficient of x^6 in the expansion $(1+x+x^2+x^3+\cdots)^n$. Use this interpretation to find this number.
23. a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1+x_2+x_3=k$ when x_1, x_2 , and x_3 are integers with $x_1 \geq 2, 0 \leq x_2 \leq 3$, and $2 \leq x_3 \leq 5$?
b) Use your answer to part (a) to find a_6 .
24. a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1+x_2+x_3+x_4=k$ when x_1, x_2, x_3 , and x_4 are integers with $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$, and $x_4 \geq 1$?
b) Use your answer to part (a) to find a_7 .
25. Explain how generating functions can be used to find the number of ways in which postage of r cents can be pasted on an envelope using 3-cent, 4-cent, and 20-cent stamps.
 - a) Assume that the order the stamps are pasted on does not matter.
 - b) Assume that the order in which the stamps are pasted on matters.
 - c) Use your answer to part (a) to determine the number of ways 46 cents of postage can be pasted on an envelope using 3-cent, 4-cent, and 20-cent stamps when the order the stamps are pasted on does not matter. (Use of a computer algebra program is advised.)
 - d) Use your answer to part (b) to determine the number of ways 46 cents of postage can be pasted in a row on an envelope using 3-cent, 4-cent, and 20-cent stamps when the order in which the stamps are pasted on matters. (Use of a computer algebra program is advised.)
26. Explain how generating functions can be used to find the number of ways in which postage of r cents can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps.
 - a) Assume that the order the stamps are pasted on does not matter.
 - b) Assume that the order the stamps are pasted on matters.
 - c) Use your answer to part (a) to determine the number of ways 49 cents of postage can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps when the order the stamps are pasted on does not matter. (Use of a computer algebra program is advised.)
 - d) Use your answer to part (b) to determine the number of ways 49 cents of postage can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps when the order in which the stamps are pasted on matters. (Use of a computer algebra program is advised.)
27. Customers at a quirky tropical fruit stand can buy at most four mangos, at most two passion fruit, any even number of papayas, three or more coconuts, and carambolas in groups of five.
 - a) Explain how generating functions can be used to find the number of ways a customer can buy n pieces of these fruits, following the restrictions listed.
 - b) Use your answer in part (a) to determine the number of ways you can buy a dozen pieces of these fruits.
28. a) Show that $1/(1-x-x^2-x^3-x^4-x^5-x^6)$ is the generating function for the number of ways that the sum n can be obtained when a die is rolled repeatedly and the order of the rolls matters.
b) Use part (a) to find the number of ways to roll a total of 8 when a die is rolled repeatedly, and the order of the rolls matters. (Use of a computer algebra package is advised.)
29. Use generating functions (and a computer algebra package, if available) to find the number of ways to make change for \$1 using
 - a) dimes and quarters.
 - b) nickels, dimes, and quarters.
 - c) pennies, dimes, and quarters.
 - d) pennies, nickels, dimes, and quarters.

30. Use generating functions (and a computer algebra package, if available) to find the number of ways to make change for \$1 using pennies, nickels, dimes, and quarters with
- no more than 10 pennies.
 - no more than 10 pennies and no more than 10 nickels.
 - * no more than 10 coins.
31. Use generating functions to find the number of ways to make change for \$100 using
- \$10, \$20, and \$50 bills.
 - \$5, \$10, \$20, and \$50 bills.
 - \$5, \$10, \$20, and \$50 bills if at least one bill of each denomination is used.
 - \$5, \$10, and \$20 bills if at least one and no more than four of each denomination is used.
32. If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for each of these sequences?
- $2a_0, 2a_1, 2a_2, 2a_3, \dots$
 - $0, a_0, a_1, a_2, a_3, \dots$ (assuming that terms follow the pattern of all but the first term)
 - $0, 0, 0, 0, a_2, a_3, \dots$ (assuming that terms follow the pattern of all but the first four terms)
 - $a_2, a_3, a_4, \dots$
 - $a_1, 2a_2, 3a_3, 4a_4, \dots$ [Hint: Calculus required here.]
 - $a_0^2, 2a_0a_1, a_1^2 + 2a_0a_2, 2a_0a_3 + 2a_1a_2, 2a_0a_4 + 2a_1a_3 + a_2^2, \dots$
33. If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for each of these sequences?
- $0, 0, 0, a_3, a_4, a_5, \dots$ (assuming that terms follow the pattern of all but the first three terms)
 - $a_0, 0, a_1, 0, a_2, 0, \dots$
 - $0, 0, 0, 0, a_0, a_1, a_2, \dots$ (assuming that terms follow the pattern of all but the first four terms)
 - $a_0, 2a_1, 4a_2, 8a_3, 16a_4, \dots$
 - $0, a_0, a_1/2, a_2/3, a_3/4, \dots$ [Hint: Calculus required here.]
 - $a_0, a_0 + a_1, a_0 + a_1 + a_2, a_0 + a_1 + a_2 + a_3, \dots$
34. Use generating functions to solve the recurrence relation $a_k = 7a_{k-1}$ with the initial condition $a_0 = 5$.
35. Use generating functions to solve the recurrence relation $a_k = 3a_{k-1} + 2$ with the initial condition $a_0 = 1$.
36. Use generating functions to solve the recurrence relation $a_k = 3a_{k-1} + 4^{k-1}$ with the initial condition $a_0 = 1$.
37. Use generating functions to solve the recurrence relation $a_k = 5a_{k-1} - 6a_{k-2}$ with initial conditions $a_0 = 6$ and $a_1 = 30$.
38. Use generating functions to solve the recurrence relation $a_k = a_{k-1} + 2a_{k-2} + 2^k$ with initial conditions $a_0 = 4$ and $a_1 = 12$.
39. Use generating functions to solve the recurrence relation $a_k = 4a_{k-1} - 4a_{k-2} + k^2$ with initial conditions $a_0 = 2$ and $a_1 = 5$.
40. Use generating functions to solve the recurrence relation $a_k = 2a_{k-1} + 3a_{k-2} + 4^k + 6$ with initial conditions $a_0 = 20, a_1 = 60$.

41. Use generating functions to find an explicit formula for the Fibonacci numbers.

*42. a) Show that if n is a positive integer, then

$$\binom{-1/2}{n} = \binom{2n}{n} / (-4)^n.$$

- b) Use the extended binomial theorem and part (a) to show that the coefficient of x^n in the expansion of $(1 - 4x)^{-1/2}$ is $\binom{2n}{n}$ for all nonnegative integers n .

*43. (Calculus required) Let $\{C_n\}$ be the sequence of Catalan numbers, that is, the solution to the recurrence relation $C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1}$ with $C_0 = C_1 = 1$ (see Example 5 in Section 8.1).

- a) Show that if $G(x)$ is the generating function for the sequence of Catalan numbers, then $xG(x)^2 - G(x) + 1 = 0$. Conclude (using the initial conditions) that $G(x) = (1 - \sqrt{1 - 4x})/(2x)$.

- b) Use Exercise 42 to conclude that

$$G(x) = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n,$$

so that

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

- c) Show that $C_n \geq 2^{n-1}$ for all positive integers n .

44. Use generating functions to prove Pascal's identity: $C(n, r) = C(n-1, r) + C(n-1, r-1)$ when n and r are positive integers with $r < n$. [Hint: Use the identity $(1+x)^n = (1+x)^{n-1} + x(1+x)^{n-1}$.]

45. Use generating functions to prove Vandermonde's identity: $C(m+n, r) = \sum_{k=0}^r C(m, r-k)C(n, k)$, whenever m, n , and r are nonnegative integers with r not exceeding either m or n . [Hint: Look at the coefficient of x^r in both sides of $(1+x)^{m+n} = (1+x)^m(1+x)^n$.]

46. This exercise shows how to use generating functions to derive a formula for the sum of the first n squares.

- a) Show that $(x^2 + x)/(1-x)^4$ is the generating function for the sequence $\{a_n\}$, where $a_n = 1^2 + 2^2 + \dots + n^2$.

- b) Use part (a) to find an explicit formula for the sum $1^2 + 2^2 + \dots + n^2$.

The **exponential generating function** for the sequence $\{a_n\}$ is the series

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n.$$

For example, the exponential generating function for the sequence $1, 1, 1, \dots$ is the function $\sum_{n=0}^{\infty} x^n/n! = e^x$. (You will find this particular series useful in these exercises.) Note that e^x is the (ordinary) generating function for the sequence $1, 1, 1/2!, 1/3!, 1/4!, \dots$.

47. Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where
- $a_n = 2$.
 - $a_n = (-1)^n$.
 - $a_n = 3^n$.
 - $a_n = n + 1$.
 - $a_n = 1/(n + 1)$.
48. Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where
- $a_n = (-2)^n$.
 - $a_n = -1$.
 - $a_n = n$.
 - $a_n = n(n - 1)$.
 - $a_n = 1/((n + 1)(n + 2))$.
49. Find the sequence with each of these functions as its exponential generating function.
- $f(x) = e^{-x}$
 - $f(x) = 3x^{2x}$
 - $f(x) = e^{3x} - 3e^{2x}$
 - $f(x) = (1 - x) + e^{-2x}$
 - $f(x) = e^{-2x} - (1/(1 - x))$
 - $f(x) = e^{-3x} - (1 + x) + (1/(1 - 2x))$
 - $f(x) = e^{x^2}$
50. Find the sequence with each of these functions as its exponential generating function.
- $f(x) = e^{3x}$
 - $f(x) = 2e^{-3x+1}$
 - $f(x) = e^{4x} + e^{-4x}$
 - $f(x) = (1 + 2x) + e^{3x}$
 - $f(x) = e^x - (1/(1 + x))$
 - $f(x) = xe^x$
 - $f(x) = e^{x^3}$
51. A coding system encodes messages using strings of octal (base 8) digits. A codeword is considered valid if and only if it contains an even number of 7s.
- Find a linear nonhomogeneous recurrence relation for the number of valid codewords of length n . What are the initial conditions?
 - Solve this recurrence relation using Theorem 6 in Section 8.2.
 - Solve this recurrence relation using generating functions.
- *52. A coding system encodes messages using strings of base 4 digits (that is, digits from the set $\{0, 1, 2, 3\}$). A codeword is valid if and only if it contains an even number of 0s and an even number of 1s. Let a_n equal the number of valid codewords of length n . Furthermore, let b_n , c_n , and d_n equal the number of strings of base 4 digits of length n with an even number of 0s and an odd number of 1s, with an odd number of 0s and an even number of 1s, and with an odd number of 0s and an odd number of 1s, respectively.
- Show that $d_n = 4^n - a_n - b_n - c_n$. Use this to show that $a_{n+1} = 2a_n + b_n + c_n$, $b_{n+1} = b_n - c_n + 4^n$, and $c_{n+1} = c_n - b_n + 4^n$.
 - What are a_1 , b_1 , c_1 , and d_1 ?
 - Use parts (a) and (b) to find a_3 , b_3 , c_3 , and d_3 .
 - Use the recurrence relations in part (a), together with the initial conditions in part (b), to set up three equations relating the generating functions $A(x)$, $B(x)$, and $C(x)$ for the sequences $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$, respectively.
 - Solve the system of equations from part (d) to get explicit formulae for $A(x)$, $B(x)$, and $C(x)$ and use these to get explicit formulae for a_n , b_n , c_n , and d_n .

Generating functions are useful in studying the number of different types of partitions of an integer n . A **partition** of a positive integer is a way to write this integer as the sum of positive integers where repetition is allowed and the order of the integers in the sum does not matter. For example, the partitions of 5 (with no restrictions) are $1 + 1 + 1 + 1 + 1$, $1 + 1 + 1 + 2$, $1 + 1 + 3$, $1 + 2 + 2$, $1 + 4$, $2 + 3$, and 5. Exercises 53–58 illustrate some of these uses.

53. Show that the coefficient $p(n)$ of x^n in the formal power series expansion of $1/((1-x)(1-x^2)(1-x^3)\cdots)$ equals the number of partitions of n .
54. Show that the coefficient $p_o(n)$ of x^n in the formal power series expansion of $1/((1-x)(1-x^3)(1-x^5)\cdots)$ equals the number of partitions of n into odd integers, that is, the number of ways to write n as the sum of odd positive integers, where the order does not matter and repetitions are allowed.
55. Show that the coefficient $p_d(n)$ of x^n in the formal power series expansion of $(1+x)(1+x^2)(1+x^3)\cdots$ equals the number of partitions of n into distinct parts, that is, the number of ways to write n as the sum of positive integers, where the order does not matter but no repetitions are allowed.
56. Find $p_o(n)$, the number of partitions of n into odd parts with repetitions allowed, and $p_d(n)$, the number of partitions of n into distinct parts, for $1 \leq n \leq 8$, by writing each partition of each type for each integer.
57. Show that if n is a positive integer, then the number of partitions of n into distinct parts equals the number of partitions of n into odd parts with repetitions allowed; that is, $p_o(n) = p_d(n)$. [Hint: Show that the generating functions for $p_o(n)$ and $p_d(n)$ are equal.]
- **58. (Requires calculus) Use the generating function of $p(n)$ to show that $p(n) \leq e^{C\sqrt{n}}$ for some constant C . [Hardy and Ramanujan showed that $p(n) \sim e^{\pi\sqrt{2/3}\sqrt{n}}/(4\sqrt{3}n)$, which means that the ratio of $p(n)$ and the right-hand side approaches 1 as n approaches infinity.]

Suppose that X is a random variable on a sample space S such that $X(s)$ is a nonnegative integer for all $s \in S$. The **probability generating function** for X is

Links 
$$G_X(x) = \sum_{k=0}^{\infty} p(X(s) = k)x^k.$$

59. (Requires calculus) Show that if G_X is the probability generating function for a random variable X such that $X(s)$ is a nonnegative integer for all $s \in S$, then
- $G_X(1) = 1$.
 - $E(X) = G'_X(1)$.
 - $V(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2$.
60. Let X be the random variable whose value is n if the first success occurs on the n th trial when independent Bernoulli trials are performed, each with probability of success p .
- Find a closed formula for the probability generating function G_X .
 - Find the expected value and the variance of X using Exercise 59 and the closed form for the probability generating function found in part (a).